

3. Three-day basketball focus

Audience: Charles — deadline pressure
Goal: One clear “done” for the simplified model, without scope creep.
Status (2026-07-27): Re-entry bar **reached** on disk (agent/project). **Your** step-by-step checkoff: CHARLES_CHECKLIST.md .

What “done” means (72-hour bar)

Binding check: You can say in one breath: *Hero = outcome*; **environment** = `L_net` ($B - D$); *advancement = score* (`S_i`, λ) then **select** (top K); *sim tests congestion in the score*, not one giant environment model. (`../BINDING_Selection_is_its_own_step.md`)

You can stop spiraling when you can honestly say all four:

- I can explain the hero** in one minute (doc 01) — including **talent baseline** vs **hero** (two empirical axes).
- I can explain three layers** without merging them — score `S_i = A_i + λ(B-D)`, Alex v1 $(B-D)=-L_C$, knockout $\lambda=0$, then **select** top K (doc 02).
- I have a side-by-side figure:** empirical hero + generative sim, same advancement rate on Y , labeled axes.
- I have one limitation sentence** ready for Alex (below — copy when needed).

You do **not** need: bin-for-bin sim match, Rivanna sweeps, p calibrated to 530, tenure figures, manuscript Word inking, or refreshed Army plots in this window.

Minimal model for Alex’s intent (v1 — what he asked for)

Piece	What it is	Status
Empirical phenomenon	Naïve: $P(\text{draft} \mid \text{own } A_i) \approx \text{monotone}$; Hero: $P(\text{draft} \mid \text{PoolQ_LOO})$ with tail dip	Done (530; doc 01)
Three-step generative	(1) assign (p); (2) score <code>S_i</code> ; (3) select top K	<code>540_*</code> + <code>Model.pdf</code>
Unified score	<code>S_i = A_i + λ(B-D)</code> ; Alex v1: $(B-D)=-L_C \Rightarrow S_i = A_i - \lambda \cdot L_C$	Locked (doc 02)
Headline sim test	Knockout $\lambda=0$ ($S_i = A_i$) vs congestion in score — same league, same top- K	Done (<code>hero_model_reset_bundle.py</code>)
Side-by-side + limits	Empirical + generative PNG; no bin-for-bin claim	Done (export folder)
p ablation	Vary assignment assortativity, fix score + winner rule — “is sorting involved?”	Pass B — <code>540_rho_ablation_bundle.py</code>

One line for Alex: *Hero is the stylized fact; the minimal generative proof is congestion in the score* (Pass A knockout done); p ablation tests whether assignment sorting moves the readout with score and top- K held fixed.

Limitation sentence (v1 — use as-is or lightly edit)

*The empirical stylized fact uses leave-one-out teammate quality among college players; the generative proof-of-concept **scores** players with a penalty for leave-one-out viable-peer congestion, then **selects** top K , and plots binned selection rates on a matching quality axis. We do not claim bin-for-bin reproduction of every ventile in v1; we claim that talent-only **scoring** is insufficient and that congestion in the **score** can bend advancement curves in a disciplined artificial league.*

What is already built (do not rebuild from scratch)

Piece	Status	Where
Empirical hero (Layer A)	Done	<code>530</code> ; <code>sports/datasets/mbb/exports_inverted_u_v0/alex_side_by_side_v0/</code>
Quadratic LPM ($\beta_2 < 0$)	Done	<code>lpm_hero_coefficients.txt</code> in export folder
Generative knockouts ($\lambda=0$ vs congestion)	Done	<code>generative_knockout*_16quantile.csv</code> ; script <code>sports/scripts/hero_model_reset_bundle.py</code>
Side-by-side PNG	Done	<code>inverted_u_side_by_side_empirical_vs_generative.png</code>
Narrative slide	Done	<code>Model.pdf</code> / <code>Model.pptx</code>
Plain readout	Done	<code>generative_knockout_summary.txt</code> , <code>side_by_side_caption.txt</code>

Re-run bundle if needed: `python sports/scripts/hero_model_reset_bundle.py`

Five-minute self-test (you’re done when these pass)

- Hero (60 s):** Naïve vs Hero; tail dip is **outcome on pool axis**, not “NBA ignores talent.”
- Binding (20 s):** environment $\neq$ advancement; score $\neq$ select (doc 03 or `Model.pdf` opener).
- Knockout (20 s):** same league + same top- K ; $\lambda=0 \rightarrow S_i=A_i$; else $S_i=A_i-\lambda L_C$ with $(B-D)=-L_C$.

- 4. **Open side-by-side PNG** — left empirical, right generative; one honest limit.
- 5. **Read limitation sentence** once without hedging.

Suggested schedule (if still orienting)

Day 1 — Layer A

- Empirical hero PNG + doc 01; describe tail dip and **talent baseline**.

Day 2 — Layer C (score knockout)

- Side-by-side PNG + knockout CSVs; λ toggles **L_C** in the score, not **p** and not the winner rule.

Day 3 — Package for Alex

- **Model.pdf** + side-by-side + limitation sentence (email or brief).

Pass B: p sims (assignment ablation)

Question: Is assortative grouping (**p**) involved? **Answer:** Yes in the **full** story (step 1 — who lands where); **not** the minimal proof that congestion in the **score** matters (Pass A — λ / **L_C**, already run).

Run: hold **score and winner rule** fixed (λ , **L_C**, top K); vary **p** (low vs high) + sort-and-chop benchmark; same 16-bin readout on **poolq_loo**.
Script: `python sports/scripts/540_rho_ablation_bundle.py`. See `../../sports/540_READ_ME_SIM.md`.

Explicitly out of scope (park guilt here)

- Bin-for-bin replication of hero ventiles from simulation
- Separate estimation of benefit vs congestion on one axis
- **p** calibrated to match hero before minimal story is sent
- Multi-domain Λ sweeps (Army slot capacity)
- Tenure Cox / Setting 3 prose
- Re-reading agent correspondence rounds

If you get pulled into cross-references again

Rule: Close the tab. Return to `00_READ_ME_FIRST.md`.

Agents: when Charles is in re-entry, **do not** point him at `Charles_reading_list.md` items 1–14, Pertinent Thoughts, or nesting note until he finishes re_entry 01–03.

One paragraph to end the reset

You started over because the **reference surface** exceeded your **working memory**. The science for Alex v1 is **on disk** and **in your slide**. Remaining work is **Pass B p packaging** (optional with Pass A) and Alex presentation — not re-deriving the model. Everything else waits in `PARKED_FOR_LATER.md`.

When you want depth beyond the slide, see `3-Master_Plan/plans/20260721_hero_model_reset.plan.md` — not before doc 02 feels easy.